

# Technology Stack and Architecture

Standard laptop order automation with Flow Designer

|                |                                    |
|----------------|------------------------------------|
| • Date         | • 31 January 3035                  |
| • Team ID      | • SWTID-2026-7609                  |
| • Project Name | • Standard laptop order automation |

## Purpose and architecture boundary

Use this worksheet to document the target technology stack, interfaces, deployment boundaries, and operational characteristics for automating standard laptop orders. Flow Designer is the requested workflow direction; the platform instance, version, features, connectors, and all adjacent systems must be confirmed by the architecture and service owners. The architecture view below is conceptual, not an approved design.

## Conceptual architecture canvas

| Actor / external system             | Input or data flow                                   | Workflow boundary                                                          | Output or data flow                         | Destination / owner                              |
|-------------------------------------|------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------|
| Employee / manager                  | Request, laptop choice, need and delivery details -> | REQUEST EXPERIENCE + FLOW DESIGNER<br>Platform / instance / version: [TBD] | Submission result, status, action needed -> | Employee / manager                               |
| Identity / worker data source [TBD] | Worker and organizational context ->                 | Catalog checks, request validation, policy-based routing [confirm rules]   | Approval task / recorded decision <->       | Authorized approver / manager                    |
| Catalog / policy / inventory [TBD]  | Approved choices, eligibility, availability <->      | Flow orchestration and exception handling                                  | Approved order request / reference <->      | Procurement or purchasing system [TBD]           |
| Procurement / fulfillment [TBD]     | Order progress, receipt, delivery <->                | Status coordination and audit trail                                        | Asset / setup handoff, requester updates -> | IT asset / endpoint / notification systems [TBD] |

Arrow key: -> sends data to the next element; <- returns data to the previous element; <-> indicates a two-way exchange. Replace conceptual labels with confirmed system names, boundaries, data ownership, and interface direction.

## Architecture decisions to capture

| Boundary / decision                            | Record                                                        |
|------------------------------------------------|---------------------------------------------------------------|
| Included actors, process start/end, exclusions | [Define standard request boundary and exception workflows]    |
| System of record for each data set             | [Name source, owner, and authoritative fields]                |
| Hosting, network zones, environments           | [Development / test / production boundaries and connectivity] |



# Table 1 Components and technologies

Record the selected product, service, version, configuration, environment, and owner. “TBD” means a decision is still open; examples in the component column are categories, not mandated products.

| No. | Component                        | Purpose / data responsibility                                                | Selected technology / product            | Version / configuration / environment   | Owner / status   |
|-----|----------------------------------|------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------|------------------|
| 01  | Requester experience / catalog   | Collect standard laptop choice, request details, and status.                 | [Service catalog / portal / channel TBD] | [Instance, version, config TBD]         | [Owner / status] |
| 02  | Workflow orchestration           | Coordinate validation, approvals, handoffs, status, and exceptions.          | Flow Designer [platform TBD]             | [Instance, version, flow ownership TBD] | [Owner / status] |
| 03  | Identity and worker context      | Provide approved requester, role, location, manager, or cost context.        | [Directory / HR source TBD]              | [Interface, fields, refresh TBD]        | [Owner / status] |
| 04  | Laptop catalog and policy        | Define standard configurations, eligibility, and approval rules.             | [Catalog / policy source TBD]            | [Rule owner and version TBD]            | [Owner / status] |
| 05  | Approval tasking                 | Assign policy-required decisions and record outcomes.                        | [Workflow approval capability TBD]       | [Delegation, timeout, audit TBD]        | [Owner / status] |
| 06  | Procurement / purchasing         | Create or hand off approved purchase information and return order reference. | [ERP / procurement system TBD]           | [Interface, order fields TBD]           | [Owner / status] |
| 07  | Inventory / supplier fulfillment | Check availability and receive order / shipment milestones.                  | [Inventory / supplier source TBD]        | [Availability source and cadence TBD]   | [Owner / status] |
| 08  | Asset / endpoint operations      | Capture receipt, serial, assignment, setup, and handoff details.             | [Asset / endpoint system TBD]            | [System of record and lifecycle TBD]    | [Owner / status] |
| 09  | Notifications and status         | Communicate agreed milestones and actions to intended users.                 | [Email / collaboration / portal TBD]     | [Channels and templates TBD]            | [Owner / status] |
| 10  | Request and audit data           | Retain request state, decisions, references, and history as approved.        | [Platform data store / audit source TBD] | [Access, retention, classification TBD] | [Owner / status] |
| 11  | Integration / connectivity       | Connect approved systems and manage authentication and errors.               | [Native action / API / middleware TBD]   | [Network path, auth, retry TBD]         | [Owner / status] |
| 12  | Runtime and operations           | Host, monitor, support, promote, and recover the solution.                   | [Platform hosting / environments TBD]    | [Deployment, monitoring, support TBD]   | [Owner / status] |

## Table 2 Interfaces and data boundaries

Complete one row per interface. Confirm direction, data fields, authentication, network path, failure behavior, and data classification with both system owners.

| ID     | Interface / endpoints                      | Data exchanged                                     | Direction / trigger / cadence  | Security / zone                      | Owner / error handling                  |
|--------|--------------------------------------------|----------------------------------------------------|--------------------------------|--------------------------------------|-----------------------------------------|
| INT-01 | Flow Designer <-> identity / worker source | [Requester, manager, role, location, cost fields]  | [Read / trigger / cadence TBD] | [Auth, network zone TBD]             | [Owner, failure path TBD]               |
| INT-02 | Flow Designer <-> catalog / policy         | [Models, eligibility, policy version]              | [Read / update / source TBD]   | [Access and change control TBD]      | [Owner, freshness TBD]                  |
| INT-03 | Flow Designer <-> approval tasking         | [Approval context, decision, reason, timestamp]    | [Create / update / return TBD] | [Approver access / audit TBD]        | [Owner, timeout behavior TBD]           |
| INT-04 | Flow Designer <-> procurement              | [Approved configuration, cost, delivery, order ID] | [Submit / status return TBD]   | [Authentication, network path TBD]   | [Owner, retry / duplicate handling TBD] |
| INT-05 | Flow Designer <-> inventory / supplier     | [Availability, lead time, shipment, receipt]       | [Query / event / batch TBD]    | [Supplier boundary and security TBD] | [Owner, stale-data behavior TBD]        |
| INT-06 | Flow Designer <-> asset / endpoint         | [Serial, asset ID, assignee, setup/handoff state]  | [Create / update / return TBD] | [Minimum data and access TBD]        | [Owner, reconciliation TBD]             |
| INT-07 | Workflow -> notification channel           | [Milestone, action required, safe message content] | [Event / notification TBD]     | [Delivery and privacy controls TBD]  | [Owner, bounce / failure path TBD]      |

## Environment and deployment plan

| Environment       | Purpose / users                           | Configuration and data separation                       | Promotion / access controls                 | Owner / status |
|-------------------|-------------------------------------------|---------------------------------------------------------|---------------------------------------------|----------------|
| Development       | [Build and unit validation]               | [Synthetic or approved test data; secrets handling TBD] | [Access, change control, deploy method TBD] | [TBD]          |
| Test / validation | [Integration, acceptance, security tests] | [Masked / test data; connected systems TBD]             | [Approvals, test evidence, rollback TBD]    | [TBD]          |
| Production        | [Live standard laptop requests]           | [Production data, retention, monitoring TBD]            | [Release authority, rollback, support TBD]  | [TBD]          |

## Cross-boundary checks

- Identify who can see or change request, employee, purchasing, approval, and asset data at each hop.
- Document secrets, certificates, credentials, network paths, logging, retention, and supplier access with security owners.
- Define behavior for stale source data, unavailable integrations, duplicates, partial completion, and manual recovery.

# Application characteristics and technical constraints

Set values with owners. Avoid “fast”, “secure”, or “highly available” without a measurable target and a verification method.

| Characteristic                  | Requirement / target to define                                                           | Verification                                | Owner / status       |
|---------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------|----------------------|
| Security and identity           | [Authentication, role access, least privilege, service identities, separation of duties] | [Threat / access test]                      | [Owner / target TBD] |
| Privacy and data lifecycle      | [Minimum necessary fields, classification, retention, deletion, permitted use]           | [Privacy / data review]                     | [Owner / target TBD] |
| Audit and traceability          | [Decision history, actor, timestamps, request/order references, retention]               | [End-to-end trace test]                     | [Owner / target TBD] |
| Performance                     | [Response / processing targets by workflow stage and request volume]                     | [Load / timing test; metric and percentile] | [Owner / target TBD] |
| Availability and support        | [Service hours, availability goal, maintenance, support response]                        | [Monitoring / support readiness]            | [Owner / target TBD] |
| Reliability and recovery        | [Timeouts, retries, idempotency, duplicate handling, manual recovery]                    | [Failure / recovery test]                   | [Owner / target TBD] |
| Usability and accessibility     | [Clear status and errors, keyboard / assistive technology, supported devices]            | [Usability / accessibility review]          | [Owner / target TBD] |
| Integration quality             | [Field mapping, freshness, reconciliation, contract version, error alerting]             | [Contract / reconciliation tests]           | [Owner / target TBD] |
| Scalability and maintainability | [Request volume, flow ownership, versioning, change and release controls]                | [Capacity / operational review]             | [Owner / target TBD] |
| Observability                   | [Logs, dashboards, alert thresholds, runbooks, incident ownership]                       | [Operational acceptance]                    | [Owner / target TBD] |

## Technology decisions and risks

| Decision / risk                                                      | Options or evidence needed                              | Decision owner | Status / due |
|----------------------------------------------------------------------|---------------------------------------------------------|----------------|--------------|
| Platform instance, version, and supported Flow Designer capabilities | [Confirm target environment and platform guardrails]    | [TBD]          | [Open]       |
| Authoritative systems and integration pattern                        | [Native actions, APIs, middleware, network constraints] | [TBD]          | [Open]       |
| Purchase / inventory / asset data ownership                          | [System of record, required fields, data freshness]     | [TBD]          | [Open]       |
| Operational ownership and recovery                                   | [Flow support, retries, alerts, manual fallback]        | [TBD]          | [Open]       |

## Review and approval

| Role                                   | Name / decision | Date / status |
|----------------------------------------|-----------------|---------------|
| Solution / platform architecture owner | [TBD]           | [TBD]         |
| IT procurement / process owner         | [TBD]           | [TBD]         |
| Security / privacy / operations        | [TBD]           | [TBD]         |